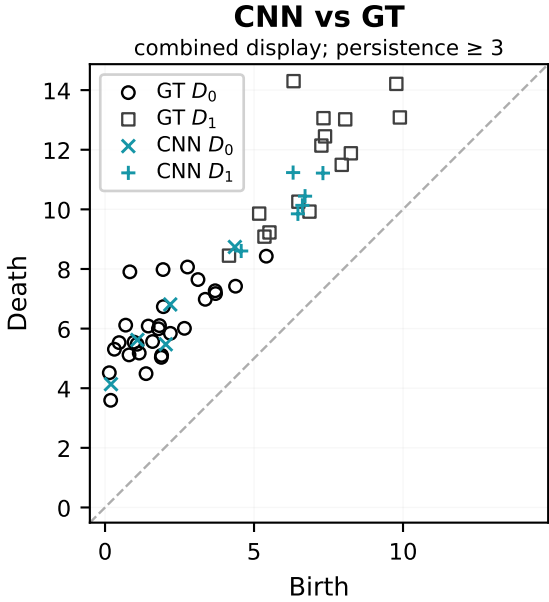
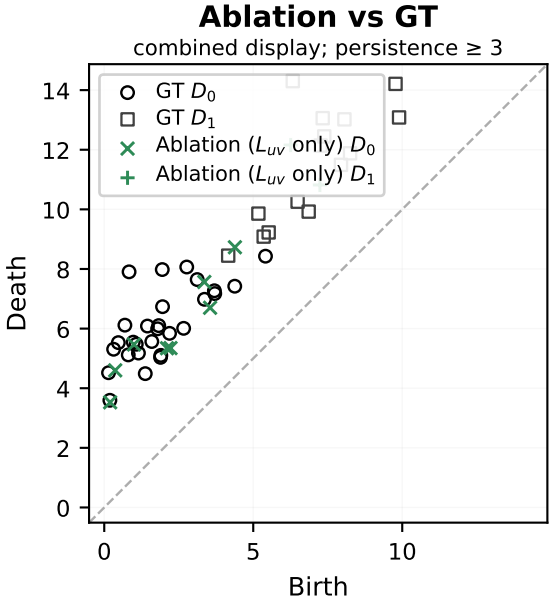
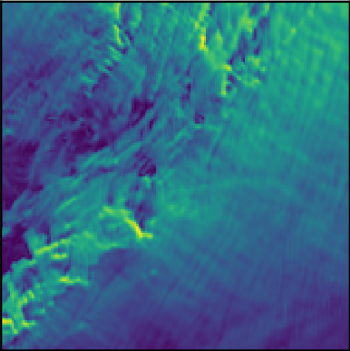


Sample 69: Topology-Aware Gains Beyond Matched Fine-Tuning

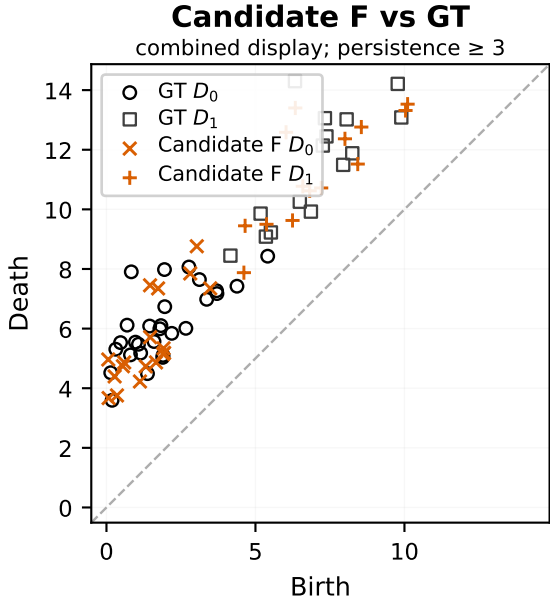
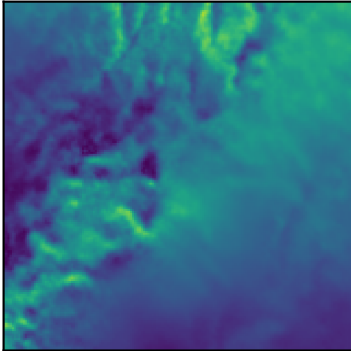
CNN = pretrained baseline; Ablation = matched L_{uv} -only fine-tuning; Candidate F = $L_{uv} + L_{grad}$ + repaired E2 supervision.



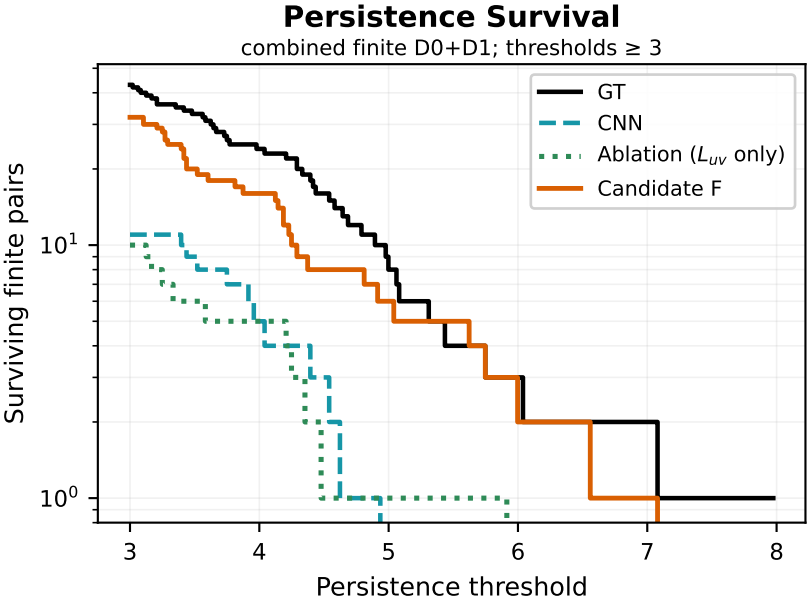
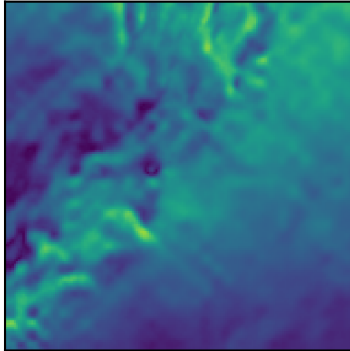
GT



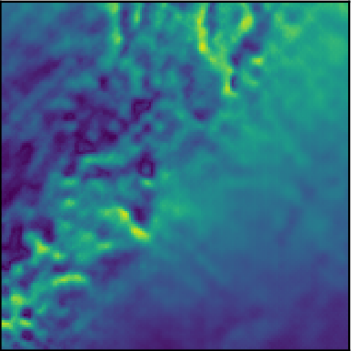
CNN



Ablation

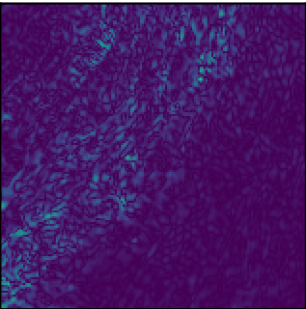


Candidate F

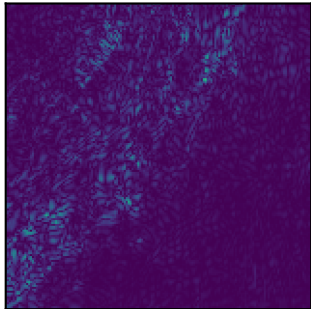


Wind-speed magnitude

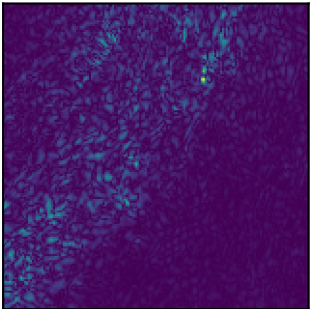
|CNN - GT|



|Ablation - GT|



|Candidate F - GT|



Absolute speed error

Topology-aware (lower is better)

$W_{2,\infty}$: CNN 14.572 $\rightarrow$ Abl. 15.256 $\rightarrow$ F 9.987
 $W_{2,2}$: CNN 18.722 $\rightarrow$ Abl. 19.913 $\rightarrow$ F 12.323

Conventional fidelity gaps

F vs CNN (original closeness rank 27/168): $|\Delta\text{PSNR}|=0.775$ dB, MAE=9.5%, RMSE=4.7%
F vs L_{uv} ablation: $|\Delta\text{PSNR}|=1.277$ dB, MAE=17.2%, RMSE=16.3%

Matched-control interpretation

Candidate F is lower than the matched L_{uv} -only fine-tuning control under all three validated PD metrics. This sample therefore supports an auxiliary-loss effect beyond ordinary fine-tuning.